# 第三卷活页选择题和填空题答案

练习一

1-1 B

1-2 D

1-3 A

1-4 A

1-5 D

1-6 $8.1\times10^{-8}/(1.67\times10^{-26})=4.85\times10^{18}$

1-7 $\dfrac{2qy\hat{j}}{4\pi\varepsilon_0(a^2+y^2)^{3/2}}$； $\pm a/\sqrt{2}$

1-8 $\lambda_{e1}d/(\lambda_{e1}+\lambda_{e2})$

练习二

2-1 C

2-2 A

2-3 A

2-4 C

2-5 D

2-6 $\lambda_e/\varepsilon_0$

2-7 $\lambda_e d/\varepsilon_0$； $\dfrac{\lambda d}{\pi\varepsilon_0(4R^2-d^2)}$； 沿矢径方向

2-8 $-\dfrac{3\sigma_e}{2\varepsilon_0}$； $-\dfrac{\sigma_e}{2\varepsilon_0}$； $\dfrac{\sigma_e}{2\varepsilon_0}$； $\dfrac{3\sigma_e}{2\varepsilon_0}$

练习三

3-1 C

3-2 B

3-3 B

3-4 D

3-5 C

3-6 $\dfrac{\lambda_e d}{4\pi\varepsilon_0}\ln\dfrac{3}{4}$；0

3-7 $\dfrac{U_{12}R_1R_2}{(R_2-R_1)r^2}\hat{e}_r$

3-8 $\dfrac{Q}{4\pi\varepsilon_0 R}\left(1-\dfrac{\Delta S}{4\pi R^2}\right)$

3-9 $-2pE\cos\alpha$

练习四

4-1 C

4-2 C

4-3 D

4-4 B

4-5 C

4-6 $-q$；球壳外的整个空间

4-7 $\sigma_e$；$\dfrac{\sigma_e}{\varepsilon_0\varepsilon_r}$

4-8 $\lambda/(2\pi r)$；$\lambda/(2\pi\varepsilon_0\varepsilon_r r)$

练习五

5-1 D

5-2 B

5-3 D

5-4 C

5-5 D

5-6（1） $D_1=0$；$D_2=\dfrac{Q}{4\pi r_2^2}$；$D_3=0$；$D_4=\dfrac{Q}{4\pi r_4^2}$；

（2） $E_1=0$；$E_2=\dfrac{Q}{4\pi\varepsilon_0 r_2^2}$；$E_3=\dfrac{Q}{4\pi\varepsilon_0\varepsilon_r r_3^2}$；$E_2=\dfrac{Q}{4\pi\varepsilon_0 r_4^2}$；

$$\varphi=\frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{R_0}-\frac{1}{R_1}+\frac{1}{\varepsilon_r R_1}-\frac{1}{\varepsilon_r R_2}+\frac{1}{R_2}\right)$$

5-7 $\varepsilon_0\varepsilon_r U^2/(2d^2)$

5-8 $\vec{E}=\begin{cases}-nq\vec{x}/\varepsilon_0 & (-d\le x\le 0)\\ nq\vec{x}/\varepsilon_0 & (0\le x\le d)\end{cases}$； $\varphi=\dfrac{nq}{2\varepsilon_0}(d^2-x^2)\quad(-d\le x\le d)$

练习六

6-1 C

6-2 A

6-3 D

6-4 D

6-5 C

6-6 $\dfrac{\mu_0\lambda\omega}{4\pi}\ln\dfrac{a+b}{a}$

6-7 $1.4\times10^{-5}$ T

6-8 $1.72\times10^{9}$ A，自东向西

练习七

7-1 D

7-2 B

7-3 C

7-4 B

7-5 A

7-6 $\pi R^2 c$

7-7 $BIR$，垂直于纸面向里

7-8 $I\cdot\tan\theta$

练习八

8-1 A

8-2 B

8-3 D

8-4 D

8-5 C

8-6 $v=\dfrac{eBR}{m}=\left(\sqrt{2}+1\right)\dfrac{leB}{m}$

8-7 $R_A:R_B=2$，$T_A:T_B=1$

8-8 增加

练习九

9-1 C

9-2 A

9-3 B

9-4 C

9-5 D

9-6　一个电源；$vBL$；洛伦兹力

9-7 $\dfrac{\mu_0 I\pi r^2}{2a}\cos\omega t$，$\dfrac{\mu_0 I\omega\pi r^2}{2Ra}\sin\omega t$

9-8 随时间变化的磁场；无头无尾的封闭曲线

练习十

10-1 A

10-2 C

10-3 B

10-4 A

10-5 C

10-6 0，$\dfrac{2\mu_0 I^2}{9\pi^2 d^2}$

10-7　$\mathrm{d}\Phi_D/\mathrm{d}t=\varepsilon_0\pi R^2(\mathrm{d}E/\mathrm{d}t)$

10-8　②；③；①